

U.G.C. NET Exam

COMPUTER

(Solved with Explanation)

1. The three outputs $x_1x_2x_3$ from the 8×3 priority encoder are used to provide a vector address of the form $101x_1x_2x_300$. What is the second highest priority vector address in hexadecimal if the vector addresses are starting from the one with the highest priority?
- (a) BC (b) A4
(c) BD (d) AC

Ans : (b) Priority encoder 8×3 gives equivalent bits for digits from 0–7

Given vector address $101x_1x_2x_300$, first highest priority $101\ 0\ 00\ 00$ i.e. A0 (Since vector address are starting from one with the highest priority $x_1 = x_2 = x_3 = 0$)

Second highest priority 10100100 i.e. (A4)₁₆

($x_1 = x_2 = 0, x_3 = 1$)

So answer will be option (b)

2. What will be the output at PORT1 if the following program is executed?
- ```
MVI B, 82H
MOV A, B
MOV C, A
MVI D, 27H
OUT PORT1
HLT
```

- (a) 37H (b) 82H  
(c) B9H (d) 00H

**Ans :** (b) MVI B, 82H || move immediate 82 in to register B.

MOV A, B || move value of B to accumulator A

MOV C, A || move value A to register C

MVI D, 27H || copy value 27H to register D

OUT PORT1 || copy value of A(82H) to port 1

So output will be 82H

3. Which of the following 8085 microprocessor hardware interrupt has the lowest priority?
- (a) RST 6.5 (b) RST 7.5  
(c) TRAP (d) INTR

**Ans :** (d) Hardware interrupts in 8085.

TRAP RST 4.5 > RST 7.5 > RST 6.5 > RST 5.5 > INTR.

INTR is level triggered and maskable interrupt. It has lowest priority.

4. A dynamic RAM is refresh cycle of 32 times per msec. Each refresh operation requires 100 nsec and a memory cycle requires 250 nsec. What percentage of memory's total operating time is required for refreshes?

n. December, 2015

## R SCIENCE

### anation Paper–III)

(a) 0.64

(b) 0.96

(c) 2.00

(d) 0.32

**Ans :** (d) Given that in 1 msec ( $10^6$  nsec) no. of refresh cycle = 32 cycle then we can find cycle in 250 nsec (which is 1 memory cycle no. of refresh cycle =  $(32 * 250) 10^6$  cycle.

Now we can find total time taken by refresh operation in 1 memory cycle.

1 cycle of refresh operation required 100 nsec

$(32 * 250) / 10^6$  cycle of refresh operation required =  $((32 * 250) / (10^6)) * 100 = 0.8$  nsec

Now find percentage of memory total operating time is required for refreshes  $(100 / 250) * 0.8 = 0.32\%$

**5. A DMA controller transfers 32-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 4800 characters per second. The CPU is fetching and executing instructions at an average rate of one million instructions per second. By how much will the CPU be slowed down because of the DMA Transfer?**

(a) 0.06%

(b) 0.12%

(c) 1.2%

(d) 2.5%

**Ans :** (b) 1 Character = 1 byte

32 bit word = 4 byte

4800 bytes in 1 sec

so for 1 byte it takes  $1/4800$  sec

so for 4 byte it takes  $1/1200$  sec

so 1200 words are transferred in 1 sec through cycle stealing & gives 1 million instructions are fetching & executing in 1 sec out of which 1200 words are executing through cycle stealing so

% of slow down or cycle wasted in dma transfer =  $(1200 / 1000000) * 100 = 0.12\%$

**6. A CPU handles interrupt by executing interrupt service subroutine .....**

(a) by checking interrupt register after execution of each instruction

(b) by checking interrupt register at the end of the fetch cycle

(c) whenever an interrupt is registered

(d) by checking interrupt register at regular time interval

**Ans :** (a) A CPU handles interrupt by executing interrupt service subroutine by checking the interrupt register after finishing the executing of the current instruction.

7. Given the following set of prolog clauses :

father (X, Y) :  
parent (X, Y),  
male (X),  
parent (Sally, Bob),  
parent (Jim, Bob),  
parent (Alice, Jane),  
parent (Thomas, Jane),  
male (Bob),  
male (Jim),  
female (Salley),  
female (Alice).

How many atoms are matched to the variable 'X' before the query father (X, Jane) reports a Result?

- (a) 1 (b) 2  
(c) 3 (d) 4

**Ans :** (\*) Father (X, Jane) is true for those values of X which make sub-goals male(X) and parents (Jane, Y, X) true. Recursively apply this reasoning until reach rules that are facts; called backward chaining only “Alice” is matched.

8. Forward chaining systems are ..... where as backward chaining systems are .....

- (a) Data driven, Data driven  
(b) Goal driven, Data driven  
(c) Data driven, Goal driven  
(d) Goal driven, Goal driven

**Ans :** (c) Since data drives forward chaining it is data driven & goal drives backward chaining it is goal driven.

9. Match the following w.r.t. programming language :

| List - I |        | List - II |                                 |
|----------|--------|-----------|---------------------------------|
| (A)      | JAVA   | (i)       | Dynamically object oriented     |
| (B)      | Python | (ii)      | Statically Non-object oriented  |
| (C)      | Prolog | (iii)     | Statically object oriented      |
| (D)      | ADA    | (iv)      | Dynamically non-object oriented |

**Codes :**

- |     |       |       |      |       |
|-----|-------|-------|------|-------|
|     | (A)   | (B)   | (C)  | (D)   |
| (a) | (iii) | (i)   | (ii) | (iv)  |
| (b) | (i)   | (iii) | (ii) | (iv)  |
| (c) | (i)   | (iii) | (iv) | (ii)  |
| (d) | (ii)  | (iv)  | (i)  | (iii) |

**Ans :** (\*)

10. The combination of an IP address and a port number is known as .....

- (a) network number  
(b) socket address  
(c) subnet mask number  
(d) MAC address

**Ans :** (b) A socket address is the combination of an IP address & a port number, much like one end of a telephone connection is the combination of a phone number and a particular extension.

**11. A network with bandwidth of 10 Mbps can pass only an average of 15,000 frames per minute with each frame carrying an average of 8,000 bits. What is the throughput of this network?**

- (a) 2 Mbps (b) 60 Mbps  
(c) 120 Mbps (d) 10 Mbps

**Ans :** (a) In data transmission, through put is the amount of data moved successfully from one place to another in a given period of time & typically measured in bits per second (bps), or in Mega bytes per second (Mbps) or giga bits per second (Gbps)

Here, through put =  $15000 \times 8000 / 60 = 2 \text{ Mbps}$

**12. Consider a subnet with 720 routers. If a three-level hierarchy is chosen with eight clusters, each containing 9 regions of 10 routers, then total number of entries in the routing table is.....**

- (a) 25 (b) 27  
(c) 53 (d) 72

**Ans :** (a) Eight clusters, each containing 9 regions of 10 routers is a 3 level hierarchy. Each router needs 10 entries for local, 8 entries for routing the other regions and for distant cluster it need 7 entries i.e. total  $10 + 8 + 7 = 25$ .

**13. In a classful addressing, the IP addresses with 0 (zero) as network number:**

- (a) refers to the current network  
(b) refers to broadcast on the local network  
(c) refers to broadcast on a distant network  
(d) refers to loopback testing

**Ans :** (a) If address belongs to any network with a host number of 0 (zero) is called the network address of the network. It does not denote any host.

Addresses with a network number of 0 are used refer to the current network, allowing a host to send to a neighbor without needing to know the network number of its local network (IP packets with destination network address of 0 are not routed).

**14. In electronic mail, which of the following protocols allows the transfer of multimedia messages?**

- (a) IMAP (b) SMTP  
(c) POP 3 (d) MIME

**Ans :** (d) Multipurpose internet mail extensions (MIME) is an internet standard that extends the format of email to support. Text in character sets other than ASCII. No-text attachments: Audio, Video, Image, Application programs etc. message bodies with multiple parts.

**15. A device is sending out data at the rate of 2000 bps. How long does it take to send a file of 1,00,000 characters?**

- (a) 50 (b) 200  
(c) 400 (d) 800

**Ans :** (c) File of 1,00,000 characters =  $8 \times 100\,000$   
= 8000000

rate = 2000 bps

1 sec = 2000 b

for 800000

then 400 sec needed

**16. In activity – Selection problem, each activity  $i$  has a start time  $s_i$  and a finish time  $f_i$  where  $s_i \leq f_i$ . Activities  $i$  and  $j$  are compatible if:**

- (a)  $s_i \geq f_j$  (b)  $s_j \geq f_i$   
(c)  $s_i \geq f_j$  or  $s_j \geq f_i$  (d)  $s_i \geq f_j$  and  $s_j \geq f_i$

**Ans :** (c) Two activities are compatible if they can be completed in same order & don't overlap in time. To complete two activities, start time of one activity must be greater than or equal to finish time of other activity.

$s_i \geq f_j$  or  $s_j \geq f_i$

**17. Given two sequences X and Y :**

$X = \langle a, b, c, b, d, a, b \rangle$

$Y = \langle b, d, c, a, b, a \rangle$

**The longest common subsequence of X and Y is:**

- (a)  $\langle b, c, a \rangle$  (b)  $\langle c, a, b \rangle$   
(c)  $\langle b, c, a, a \rangle$  (d)  $\langle b, c, b, a \rangle$

**Ans :** (d)  $X = \langle a, b, c, b, d, a, b \rangle$

$Y = \langle b, d, c, a, b, a \rangle$

$\langle b, c, b, a \rangle$

**18. If there are  $n$  integers to sort, each integer has  $d$  digits and each digit is in the set  $\{1, 2, \dots, k\}$ , radix sort can sort the numbers in :**

- (a)  $O(d \cdot n \cdot k)$  (b)  $O(d \cdot n^k)$   
(c)  $O((d + n)k)$  (d)  $O(d(n + k))$

**Ans :** (d) Lets consider there are 'k' buckets (as given)

Now, for each digit you repeat the following

(1) Place the digit of each number in the appropriate bin –  $O(n)$

(2) Append all the 'k' bins sequentially thus,  $d$  digit, its  $O(d(n+k))$

**19. The solution of the recurrence relation**

$$T(n) \leq \begin{cases} \theta(1) & \text{if } n \leq 80 \\ T\left(\frac{n}{s}\right) + T\left(\frac{7n}{10} + 6\right) + O(n) & \text{if } n > 80 \end{cases}$$

**is :**

- (a)  $O(\lg n)$  (b)  $O(n)$   
 (c)  $O(n \lg n)$  (d) None of the above

**Ans :** (d) Given, recurrence relation is:

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 80 \\ T(n/s) + T\left(\frac{7n}{10} + 6\right) + O(n) & \text{else} \end{cases}$$

$$T(n) = T(n/s) + T\left(\frac{7n}{10} + 6\right) + O(n)$$

$$\leq c(n/s) + c\left(\frac{7n}{10} + 6\right) + O(n)$$

$$\leq c(n/s) + \frac{7cn}{10} + 6c + O(n)$$

$$\leq c.n = \Theta(n)$$

We assume that  $S$  is constant as 5 else we can't solve recurrence relation.

Both option (b) and (c) should be true, if we assumed  $S$  is 5.

**20. Floyd-Warshall algorithm utilizes ..... to solve the all-pairs shortest paths problem on a directed graph in ..... time.**

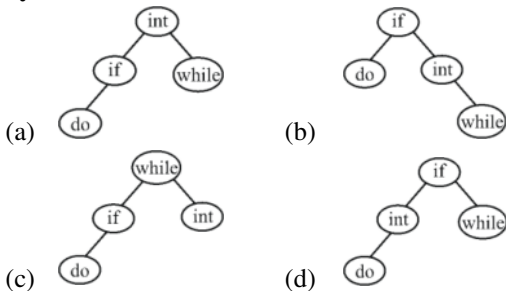
- (a) Greedy algorithm,  $\Theta(V^3)$   
 (b) Greedy algorithm,  $\Theta \log n(V^2 \lg n)$   
 (c) Dynamic programming,  $\Theta(V^3)$   
 (d) Dynamic programming,  $\Theta \log n(V^2 \lg n)$

**Ans :** (c) The Floyd -Warshall algorithm compares all possible paths through the graph between each pair of vertices. It is able to do this with  $\Theta(|V|^3)$  comparisons in a graph.

**21. Let  $n = 4$  and  $(a_1, a_2, a_3, a_4) = \text{do, if, int, while}$ . Let  $p(1 : 4) = \left(\frac{3}{8}, \frac{3}{8}, \frac{1}{8}, \frac{1}{8}\right)$  and  $q(1 : 4)$**

$$= \left(\frac{2}{8}, \frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right) \text{ where } p(i) \text{ and } q(i) \text{ denotes}$$

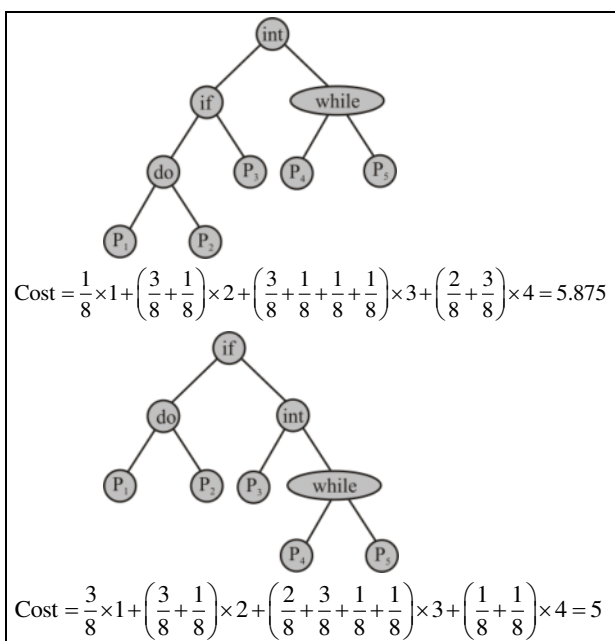
**the probability with which search  $a_i$  and the identifier  $x$  being searched satisfy  $a_i < x < a_{i+1}$  respectively. The optimal search tree is given by :**



**Ans :** (b) Search success nodes are:

$\langle \text{do, if, int, while} \rangle$  in ascending order.

Therefore, option (c) and (d) cannot be true. Now minimum cost of given tree in option (a) and (b).



22. The family of context sensitive language is ..... under union and ..... under reversal.

- (a) closed, not closed
- (b) not closed, not closed
- (c) closed, closed
- (d) not closed, closed

**Ans :** (c) Context sensitive language are closed under union, intersection, Kleen closer, concatenation and reversal.

23. Match the following :

| List – I |                                                                           | List - II |                                            |
|----------|---------------------------------------------------------------------------|-----------|--------------------------------------------|
| (A)      | $\{a^n b^n \mid n > 0\}$ is deterministic context free language           | (i)       | but not recursive language                 |
| (B)      | The complement of $\{a^n b^n a^n \mid n > 0\}$ is a context free language | (ii)      | but not context free language              |
| (C)      | $\{a^n b^n a^n\}$ is context sensitive language                           | (iii)     | but can not be accepted by a deterministic |
| (D)      | L is a recursive language                                                 | (iv)      | but not regular                            |

**Codes :**

- |     |     |      |       |       |
|-----|-----|------|-------|-------|
|     | (A) | (B)  | (C)   | (D)   |
| (a) | (i) | (ii) | (iii) | (iv)  |
| (b) | (i) | (ii) | (iv)  | (iii) |

- (c) (iv) (iii) (ii) (i)  
 (d) (iv) (iii) (i) (ii)

**Ans :** (\*) (i)  $\{a^n b^n \mid n > 0\}$  is DCFL but not regular.  
 (ii)  $\{a^n b^n a^n \mid n > 1\}$  is context sensitive language but its complement is  $\{a^m b^n a^n \mid m \neq n \text{ or } m \neq p \text{ or } n \neq p\}$  is context free language.  
 (iii) If a language is recursive, then it cannot be accepted by a deterministic push-down automata or non-deterministic push-down automata.

**24. The language of all non-null strings of a's can be defined by a context free grammar as follow:**

$$S \rightarrow aS \mid Sa \mid a$$

**The word  $a^3$  can be generated by ..... different trees.**

- (a) Two (b) Three  
 (c) Four (d) Five

**Ans :** (c)



**25. Which one of the following non-functional quality attributes is not highly affected by the architecture of the software?**

- (a) Performance (b) Reliability  
 (c) Usability (d) Portability

**Ans :** (c) Usability should not be highly affected by the architecture of the software as it is basic things rest options are extra features offered by s/w & may be highly affected by the architecture of the s/w.

**26. The context free grammar given by**

$$S \rightarrow XYZ$$

$$X \rightarrow aX \mid bX \mid A$$

$$Y \rightarrow bbb$$

**generates the language which is defined by regular expression :**

- (a)  $(a + b)^* bbb$   
 (b)  $abbb (a + b)^*$   
 (c)  $(a + b)^* (bbb) (a + b)^*$   
 (d)  $(a + b)(bbb) (a + b)^*$

**Ans :** (c) looking at  $X \rightarrow aX \mid bX \mid \epsilon$   
 we see that X is generating  $(a + b)^*$   
 Now simply replacing X by  $(a + b)^*$   
 & y by bbb is  $S \rightarrow X y X$   
 we get  $(a + b)^* (bbb) (a + b)^*$   
 So answer is option (c)

**27. There are exactly ..... different finite automata with three states x, y and z over the alphabet {a, b} where x is always the start state.**



- (a) 64 (b) 256  
(c) 1024 (d) 5832

**Ans :** (d) For  $n$  states and  $m$  input alphabets we can have number of states giving  $n^m \times 2^n$  possible DFAs when only one start state.

Given, number of states ( $n$ ) = 3

Number of input alphabets ( $m$ ) = 2

Then, number of DFA's =  $3^{2 \times 3} \times 2^3 = 3^6 \times 8 = 5832$

**28. Given the following two languages :**

$$L_1 = \{a^n b a^n \mid n > 0\}$$

$$L_2 = \{a^n b a^n b^n + 1 \mid n > 0\}$$

**Which of the following is correct?**

- (a)  $L_1$  is context free language and  $L_2$  is not context free language  
(b)  $L_1$  is not context free language and  $L_2$  is context free language  
(c) Both  $L_1$  and  $L_2$  are context free languages  
(d) Both  $L_1$  and  $L_2$  are not context free language

**Ans :** (a)  $L_1$  is context free language (deterministic) &  $L_2$  is not context free language

for  $L_1$ , we need to count the number of a's before & after b's & this length is not finite. Hence it is DCFL & not regular

for  $L_2$ , in addition to above comparison, we also need to ensure number of b's following the a's just one more.

This is a second non-finite comparison & cannot be done with just one stack. So, this makes  $L_2$  a CSL.

**29. Which of the following is used to make an Abstract class?**

- (a) Making atleast one member function as pure virtual function  
(b) Making atleast one member function as virtual function  
(c) Declaring as abstract class virtual keyword  
(d) Declaring as Abstract class static keyword

**Ans :** (a) For making the class as abstract class we should make atleast one member function as pure virtual function.

**30. Match the following with reference to object oriented modeling :**

| List – I |              | List – II |                                                                                  |
|----------|--------------|-----------|----------------------------------------------------------------------------------|
| (A)      | Polymorphism | (i)       | Picking both operator and attributes with operations appropriate model an object |
| (B)      | Inheritance  | (ii)      | Hiding implementation details of methods from users of objects                   |

|     |               |       |                                               |
|-----|---------------|-------|-----------------------------------------------|
| (C) | Encapsulation | (iii) | Using similar operations to do similar things |
| (D) | Abstraction   | (iv)  | Create new classes from existing class        |

**Codes :**

- |           |       |      |      |
|-----------|-------|------|------|
| (A)       | (B)   | (C)  | (D)  |
| (a) (iv)  | (iii) | (i)  | (ii) |
| (b) (iii) | (iv)  | (i)  | (ii) |
| (c) (iii) | (i)   | (ii) | (iv) |
| (d) (iv)  | (iii) | (ii) | (i)  |

**Ans :** (b) (a) Polymorphism is using similar operation to do similar thing (iii)

(b) Inheritance is create new class by inheriting existing one (iv)

(c) Encapsulation is showing essential feature without including background detail (i)

(d) Abstraction is (ii)

**31. In CRC based design, a CRC team consists of :**

- (1) one or two users representatives
- (2) several programmers
- (3) project co-ordinators
- (4) one or two system analysts

**Codes :**

- (a) (1) and (3)
- (b) (1), (2), (3) and (4)
- (c) (1), (3) and (4)
- (d) (1), (2) and (4)

**Ans :** (c) Answer should be C in critical review committee the CRC team must consist

(1) one or two user representative

(2) project co-ordinates

(3) system analyst

Need of programmers is not there as they can be co-ordinated later by project co-ordinators

**32. The end points of a given line are (0, 0) and (6, 18). Compute each value of y as x steps from 0 to 3, by using equation of straight line :**

- (a) For x = 0, y = 0; x = 1, y = 3; x = 2, y = 6; x = 3, y = 9
- (b) For x = 0, y = 1; x = 1, y = 3; x = 2, y = 4; x = 3, y = 9
- (c) For x = 0, y = 2; x = 1, y = 3; x = 2, y = 6; x = 3, y = 9
- (d) For x = 0, y = 0; x = 1, y = 3; x = 2, y = 4; x = 3, y = 6

**Ans :** (a) Equation of line passing through two given points  $(x_1, y_1)$  and  $(x_2, y_2)$  is—

$$\Rightarrow y - y_1 = \frac{(y_2 - y_1) \cdot (x - x_1)}{(x_2 - x_1)}$$

Given points are (0, 0) and (6, 18) then:

$$\Rightarrow y - 0 = \frac{(18 - 0) \cdot (x - 0)}{6 - 0} \Rightarrow y = 3x$$

therefore—  $x = 0 \ 1 \ 2 \ 3$

$y = 0 \ 3 \ 6 \ 9$

**33. Which of the following graphic primitives are considered as the basic building blocks of computer graphics?**

- (1) Points
- (2) Lines
- (3) Polylines
- (4) Polygons

**Codes :**

- (a) (1) only
- (b) (1) and (2)
- (c) (1), (2) and (3)
- (d) (1), (2), (3) and (4)

**Ans :** (b) A pixel is a point of light. It is just one tiny dot on the raster displays. Though it has no structure, it is definitely a building block and hence it can be considered as the graphics primitive. The resolution of CRT is related to the dot size, the diameter of single dot.

\* Line, especially straight lines, constitute the basic building block of line graphs, bar and pie charts, two and three dimensional graphs of mathematical functions, engineering drawings and architectural plans.  
– Points and lines are the building blocks of computer graphics.

**34. Javascript and Java similar name because ..... is/are true.**

- (1) Javascripts syntax is loosely based on Java's syntax
- (2) Javascript is stripped down version of Java
- (3) Java and Javascript are originated from Island of Java

**Codes :**

- (a) (1) only
- (b) (1), (2) and (3)
- (c) (1) and (2)
- (d) (1) and (3)

**Ans :** (a) Clearly statement C is not true which implies choice B, D are out also statement B is not true as they are different language.

Although they are different languages, Java Script's syntax is loosely based on Java.

**35. Which of the following statements are true with reference to the way of describing XML data?**

- (1) XML uses DTD to describe the data
- (2) XML uses XSL to describe the data
- (3) XML uses a description node to describe the data

**Codes :**

- (a) (1) only                      (b) (2) only (3)  
(c) (1) and (2)                (d) (1) and (3)

**Ans :** (d) The correct statements are (1) and (3).

Hence the answer is option (d)

**36. Which of the following is/are correct with reference to abstract class and interface?**

- (1) A class can inherit only one Abstract class but may inherit several interfaces  
(2) An Abstract class can provide complete and default code but an interface has no code

**Codes :**

- (a) (1) is true  
(b) (2) is true  
(c) Both (1) and (2) are true  
(d) Neither (1) nor (2) is true

**Ans :** (c) An interface is an empty shall, there are only the signatures of the methods, which implies that the methods do not have a body. The interface can't do anything. It's just a pattern so b is true.

A child can only extend a single class, where as an interface can extend or a class can implement multiple other interfaces so a is true.

**37. Match the following with respect to various memory management algorithms :**

| List - I |                    | List - II |                              |
|----------|--------------------|-----------|------------------------------|
| (A)      | Demand paging      | (i)       | degree of multiprogramming   |
| (B)      | Segmentation       | (ii)      | working set                  |
| (C)      | Dynamic partitions | (iii)     | supports user view of memory |
| (D)      | Fixed partitions   | (iv)      | compaction                   |

**Codes :**

- (A) (B) (C) (D)  
(a) (iii) (iv) (ii) (i)  
(b) (ii) (iii) (i) (iv)  
(c) (iv) (iii) (ii) (i)  
(d) (ii) (iii) (iv) (i)

**Ans :** (d)

**38. Function of memory management unit is :**

- (a) Address translation  
(b) Memory allocation  
(c) Cache management  
(d) All of the above

**Ans :** (a) A memory management unit (MMU), sometimes called paged memory management unit (PMMU), is a computer hardware unit having all memory references passed through itself, primarily performing the translation of virtual memory address to physical address.

**39. Consider a system with twelve magnetic tape drives and three processes  $P_1$ ,  $P_2$  and  $P_3$ . Process  $P_1$  requires maximum ten tape drives, process  $P_2$  may need as many as four tape**

drives and  $P_3$  may need upto nine tape drives. Suppose that at time  $t_1$ , process  $P_1$  is holding five tape drives, process  $P_2$  is holding two tape drives and process  $P_3$  is holding three tape drives. At time  $t_1$ , system is in :

- (a) safe state (b) unsafe state  
(c) deadlocked state (d) starvation state

**Ans :** (b) System is in UNSAFE state

- $P_1$  holds 5 tapes
- $P_2$  holds 2 tapes
- $P_3$  holds 3 tapes

Total 10 tapes are allocated &  $12 - 10 = 2$  tapes are free

- $P_1$  requires maximum 10 tape drives, it needs  $10 - 5 = 5$  more tapes
- $P_2$  may need as many as 4 tape drives, it needs  $4 - 2 = 2$  more tapes
- $P_3$  may need as many as 9 tape drives, it needs  $9 - 3 = 6$  more tapes

We can allocate 2 free tapes to  $P_2$

$P_2$  will complete execution & release all its resource including 4 tapes

Even if we allocate 4 tapes to  $P_1$  or  $P_3$  they cannot complete execution because  $P_1$  needs 5 more tape

So system is in unsafe state

**40. In Unix operating system, special files are used to :**

- (a) buffer data received in its input from where a process reads  
(b) provide a mechanism to map physical device to file names  
(c) store list of file names plus pointers associated with i-nodes  
(d) store information entered by a user application program or utility program

**Ans :** (b) In the Unix file system, six types of file are distinguished.

– special, Named pipes, Links, Symbolic links, Regular or ordinary, Directory.

\* Special- Contains no data, but provides a mechanism to map physical devices to file names. The file names are used to access peripheral devices, such as terminals and printers.

**41. Match the following in Unix file system :**

| List - I |             | List – II |                               |
|----------|-------------|-----------|-------------------------------|
| (A)      | Boot block  | (i)       | Information about file system |
| (B)      | Super block | (ii)      | Information about file        |
| (C)      | Inode table | (iii)     | Storage space                 |
| (D)      | Data block  | (iv)      | Code for making OS ready      |

**Codes :**

- |     |      |       |      |       |
|-----|------|-------|------|-------|
|     | (A)  | (B)   | (C)  | (D)   |
| (a) | (iv) | (i)   | (ii) | (iii) |
| (b) | (i)  | (iii) | (ii) | (iv)  |

- (c) (iii) (i) (ii) (iv)  
 (d) (iv) (ii) (i) (iii)

**Ans :** (a)

(a) Boot block located in the first few sectors of a file system. The boot block contains the initial bootstrap program used to load the OS (iv)

(b) A super block describes the state of the file system : the total size of the partition, the block size, pointer to a list of free blocks etc.

(c) A linear array of inodes. There is a one to one mapping of files to inode & vice versa (ii)

(d) Data blocks block containing the actual content of files (iii)

**42. In an operating system, indivisibility of operation means :**

- (a) Operation is interruptible  
 (b) Race – condition may occur  
 (c) Processor cannot be pre-empted  
 (d) All of the above

**Ans :** (c) In an OS, an divisible or irreducible operation is an atomic operation that is performed entirely or not performed at all. Means processor can not be pre-empted. It will be performed entirely or not at all.

**43. A horn clause is .....**

- (a) A clause in which no variables occur in the expression  
 (b) A clause that has at least one negative literal  
 (c) A disjunction of a number of literals  
 (d) A clause that has at most one positive literal

**Ans :** (d) In logic programming horn clause is a clause that has at most one positive literal. When there is exactly one positive literal then it is known as definite clause.

**44. In propositional logic, given P and  $P \rightarrow Q$ , we can infer .....**

- (a)  $\sim Q$  (b) Q  
 (c)  $P \wedge Q$  (d)  $\sim P \wedge Q$

**Ans :** (b)  $P \rightarrow Q$  answer P infer to Q

using modus P one ns

option (b) is the answer

**45. Reasoning strategies used in expert system include .....**

- (a) Forward chaining, backward chaining and problem reduction  
 (b) Forward chaining, backward chaining and boundary mutation  
 (c) Forward chaining, backward chaining and back propagation  
 (d) Backward chaining, problem reduction and boundary mutation

**Ans :** (a) Forward chaining : Expert system is driven by the antecedent (left hand side)

Backward chaining : Expert system is driven by the consequent (right hand side)

So answer is option (a)

**46. Language model used in LISP is .....**

- (a) Functional programming
- (b) Logic programming
- (c) Object oriented programming
- (d) All of the above

**Ans :** (a) LISP is a functional language useful properties of a functional language are :

- Evaluates an expression and use the result
- Call functions
- Related to mathematics & theory of functions
- Side effect free

**47. In constraint satisfaction problem, constraints can be stated as .....**

- (a) Arithmetic equations and inequalities that bind the values of variables
- (b) Arithmetic equations and inequalities that doesn't bind any restriction over variables
- (c) Arithmetic equations that impose restrictions over variables
- (d) Arithmetic equations that discard constraints over the given variables

**Ans :** (a) Remember map coloring problem, it is a constraint satisfaction problem we need mathematical equation and inequalities. So 3 & 4 are not the answer. Inequalities are must to check whether constraints are fullified. Equation specify the constraint. In a nutshell 1 should be the proper choice.

**48. As compared to rental and leasing methods to acquire computer systems for a management information system (MIS), purchase method has following advantage :**

- (a) It has high level of flexibility
- (b) It doesn't require cash up-front
- (c) It is a business investment
- (d) Little risk of obsolescence

**Ans :** (c) It is just an investment. Purchasing a line will not yield better reliability or flexibility than leased one.

**49. Consider the conditional entropy and mutual information for the binary symmetric channel. The input source has alphabet  $X = \{0, 1\}$  and associated probabilities  $\left\{\frac{1}{2}, \frac{1}{2}\right\}$ . The channel matrix is  $\begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}$  where  $p$  is transition probability. Then the conditional entropy is given by :**

- (a) 1
- (b)  $-p\log(p) - (1-p)\log(1-p)$
- (c)  $1 + p\log(p) + (1-p)\log(1-p)$
- (d) 0

**Ans :** (b) The uncertainty about the input  $x$  given the observed output  $y$  from the channel is the conditional entropy  $H(x/y)$ , which is defined as:

$$H(x/y) = \sum P(x, y) \log P(x/y)$$

The joint probability distribution  $P(x/y)$  is:

$$\begin{bmatrix} 0.5(1-p) & 0.5p \\ 0.5p & 0.5(1-p) \end{bmatrix}$$

and the conditional probability distribution  $P(x/y)$  is:

$$\begin{pmatrix} (1-p) & p \\ p & (1-p) \end{pmatrix}$$

Combining these matrix elements accordingly gives us the conditional entropy:

$$H(x/y) = -[0.5(1-p) \log(1-p) + 0.5p \log(p) + 0.5p \log(p) + 0.5(1-p) \log(1-p)]$$

$$H(x/y) = -(1-p) \log(1-p) - p \log(p)$$

$$H(x/y) = -p \log(p) - (1-p) \log(1-p)$$

**50. Which of the following is not a lossy compression technique?**

- (a) JPEG (b) MPEG  
(c) FFT (d) Arithmetic coding

**Ans :** (d) Arithmetic coding is a form of entropy encoding used in lossless data compression. Normally, a string of characters such as the words "Hello" is represented using a fixed number of bits per character, as in the ASCII code.

Lossless and lossy compression are terms that describe whether or not, in the compression of a file, all original data can be recovered when the file is uncompressed. Hence option (a), (b), (c) are lossy compression technique and the answer is option (d)

**51. Blind image deconvolution is .....**

- (a) Combination of blur identification and image restoration  
(b) Combination of segmentation and classification  
(c) Combination of blur and non-blur image  
(d) None of the above

**Ans :** (a) In image processing, a blind deconvolution is a deconvolution technique that permits recovery of the target scene from the single or set of "blurred" images in the presence of a poorly determined or unknown point spread function (PSF)

**52. A basic feasible solution of a linear programming problem is said to be ..... if at least one of the basic variable is zero.**

- (a) degenerate (b) non-degenerate  
(c) infeasible (d) unbounded

**Ans :** (a) Basic feasible solution : A feasible solution to LP problem which is also the basic solution is called the "basic feasible solution". Basic feasible solution are of two types.

Degenerate : A basic feasible solution is called degenerate if value of at least one basic variable is zero. So answer is option (a)



53. Consider the following conditions :

- (1) The solution must be feasible, i.e. it must satisfy all the supply and demand constraints
- (2) The number of positive allocations must be equal to  $m + n - 1$ , where  $m$  is the number of rows and  $n$  is the number of columns
- (3) All the positive allocations must be in independent positions

The initial solution of a transportation problem is said to be non-degenerate basic feasible solution if it satisfies :

Codes :

- (a) (1) and (2) only      (b) (1) and (3) only  
(c) (2) and (3) only      (d) (1), (2) and (3)

**Ans :** (d) All condition must be true (all these are basic constraints). Hence answer is option (d)

54. Consider the following transportation problem:

| Factories | Stores |    |    |     |    |        |
|-----------|--------|----|----|-----|----|--------|
|           |        | I  | II | III | IV | Supply |
|           | A      | 4  | 6  | 8   | 13 | 50     |
|           | B      | 13 | 11 | 10  | 8  | 70     |
|           | C      | 14 | 4  | 10  | 13 | 30     |
|           | D      | 9  | 11 | 13  | 8  | 50     |
|           | Demand | 25 | 35 | 105 | 20 |        |

The transportation cost in the initial basic feasible solution of the above transportation problem using Vogel's Approximation method is :

- (a) 1450      (b) 1465  
(c) 1480      (d) 1520

**Ans :** (b) Add fifth column with zero cost to balance the demand supply.

$$15 \times 0 + 30 \times 4 + 25 \times 4 + 5 \times 6 + 20 \times 8 + 20 \times 8 + 70 \times 10 + 15 \times 13 + 0 + 120 + 100 + 30 + 160 + 700 + 195 = 1465$$

55. The character set used in windows 2000 operating system is .....

- (a) 8 bit ASCII  
(b) Extended ASCII  
(c) 16 bit UNICODE  
(d) 12 bit UNICODE

**Ans :** (c) 16 bit UNICODE character set used in Windows 2000 operating system. Extended ASCII character encoding are 8 bit or larger encoding that include 7 bit ASCII character. ASCII coding is 8 bit character encoding.

56. In Unix, the command to enable execution permission for file "mylife" by all is .....

- (a) Chmod ugo + X myfile  
(b) Chmod a + X myfile  
(c) Chmod + X myfile  
(d) All of the above

**Ans :** (d) All three commands are used to enable execution permission CH mode X for execution ugo (user group other) and + for add  
a (all) and + for add  
only + for all  
So answer is option (d)

**57. What will be the output of the following Unix command?**

**\$rm chap0\[1 – 3\]**

- (a) Remove file chap0[1 – 3]
- (b) Remove file chap01, chap02, chap03
- (c) Remove file chap\[1 – 3\]
- (d) None of the above

**Ans :** (a) If you use rm chap\*, you will remove all files beginning with chap. The way to remove the file chap\* is to use the backslash (\) as follows:

**\$ chap \ \***

The use of the \ in removing any special character is called escaping or despecializing. For Ex-  
to remove a file chap 0[1 – 3], use, use: \$rm chap\*\[1 – 3\].

**58. Which of the following statement regarding the features of the object-oriented approach to databases are true?**

- (1) The ability to develop more realistic models of the real world
- (2) The ability to represent the world in a non-geometric way
- (3) The ability to develop databases using natural language approaches
- (4) The need to split objects into their component parts
- (5) The ability to develop database models based on location rather than state and behavior

**Codes :**

- (a) (1), (2) and (3)
- (b) (2), (3) and (4)
- (c) (1), (4) and (5)
- (d) (3), (4) and (5)

**Ans :** (a) The following statements are true. The ability to develop more realistic models of the real world (a)

- The ability to represent the world in a non geometric way (b)
- The ability to develop databases using natural language approach (c)

Hence answer is (a)

**59. Consider the following database table :**

**Create table test (**

**one integer,  
two integer,  
primary key (one),  
unique (two),  
check(one > = 1 and < = 10),  
check(two > = 1 and < = 5)**

**How many data records/tuples atmost can this table contain?**

- (a) 5 (b) 10  
(c) 15 (d) 50

**Ans :** (a) Attribute 'one' can have 10 values from 1 to 10 and attribute 'two' can have 5 values from 1 to 5. But attribute 'one' is primary key it cannot have duplicate values, so attribute 'one' is allowed only 5 values from 1 to 10. Any 5 value from 1 to 5 of attribute 'two' so, relation can have atmost 5 values.

**60. Suppose ORACLE relation R(A, B) currently has tuples {(1, 2), (1, 3), (3, 4)} and relation S(B, C) currently has {(2, 5), (4, 6), (7, 8)}. Consider the following two SQL queries SQ1 and SQ2 :**

**SQ1 : Select \***

**From R Full Join S**

**On R.B = S.B;**

**SQ2 : Select \***

**From R Inner Join S**

**On R.B = S.B;**

**The numbers of tuples in the result of the SQL query SQ1 and the SQL query SQ2 are given by :**

- (a) 2 and 6 respectively  
(b) 6 and 2 respectively  
(c) 2 and 4 respectively  
(d) 4 and 2 respectively

**Ans :** (d) Here  $R.B = S.B$  holds for 2 records when  $B = 2$  &  $B = 4$  hence its result will be 2

for SQ2 full join or full outer join gives the result of left outer join (all rows of left table whether there is match or not with the right) now both table has 3 records but there are 2 match and 2 mismatch.

Hence SQ1 will give 4 records

So answer is option (d)

**61. Consider the following three SQL queries (Assume the data in the people table) :**

- (1) **Select Name from people where Age > 21;**  
(2) **Select Name from people where Height > 180;**  
(3) **Select Name from people where (age > 21) or (Height > 180);**

**If the SQL queries (1) and (2) above, return 10 row and 7 rows in the result set respectively, then what is one possible number of rows returned by the SQL query (3)?**

- (a) 3 (b) 7  
(c) 10 (d) 21

**Ans :** (c) When set a and set b all rows are distinct answer would be  $= 10 + 7 = 17$

When set b is subset of set a then  $10 + 7 - 7 = 10$

So answer is option (c)

**62. The distributed system is a collection of .....(P)..... and communication is achieved in**

distributed system by .....(Q)..... where (P) and (Q) are :

- (a) Loosely coupled hardware on tightly coupled software and disk sharing, respectively.
- (b) Tightly coupled hardware on loosely coupled software and shared memory, respectively
- (c) Tightly coupled software on loosely coupled hardware and message passing, respectively hardware and file sharing, respectively
- (d) Loosely coupled software on tightly coupled hardware and file sharing, respectively

**Ans :** (c) Hardware is loosely coupled & software is tightly coupled.

Distributed computing is a field of computer science that studies distributed systems. A distributed system is model in which components located on network computers communicate & co-ordinate their actions by passing message.

**63. Consider the following three tables R, S and T. In this question, all the join operations are natural joins ( $\bowtie$ ). ( $\pi$ ) is the projection operation of a relation :**

| R |   | S |   | T |   |
|---|---|---|---|---|---|
| A | B | B | C | A | C |
| 1 | 2 | 6 | 2 | 7 | 1 |
| 3 | 2 | 2 | 4 | 1 | 2 |
| 5 | 6 | 8 | 1 | 9 | 3 |
| 7 | 8 | 8 | 3 | 5 | 4 |
| 9 | 8 | 2 | 5 | 3 | 5 |

**Possible answer tables for this are also given as below :**

| A | B | C | A | B | C | A | B | C | A | B | C |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 4 | 1 | 2 | 2 | 1 | 6 | 2 | 3 | 2 | 5 |
| 1 | 2 | 5 | 3 | 2 | 5 | 3 | 2 | 5 | 7 | 8 | 1 |
| 3 | 2 | 4 | 5 | 6 | 4 | 5 | 2 | 4 | 7 | 8 | 1 |
| 3 | 2 | 5 | 7 | 8 | 1 | 7 | 8 | 1 | 9 | 8 | 3 |
| 5 | 6 | 2 | 9 | 8 | 3 | 9 | 8 | 3 | 9 | 8 | 3 |
| 7 | 8 | 1 |   |   |   |   |   |   |   |   |   |
| 7 | 8 | 3 |   |   |   |   |   |   |   |   |   |
| 9 | 8 | 1 |   |   |   |   |   |   |   |   |   |
| 9 | 8 | 3 |   |   |   |   |   |   |   |   |   |

**What is the resulting table of  $\pi_{A,B}(R \bowtie T) \pi_{A,B}(R \bowtie T) \bowtie \pi_{B,C}(S \bowtie T)$ ?**

- (a) (1)
- (b) (2)
- (c) (3)
- (d) (4)

**Ans :** (b) Option (a) is option return all possible results after natural join but it does not involve the table t during natural join. So option (b) is correct.

**64. Consider the two class classification task that consists of the following points :**

**Class C1 :**  $[-1, -1], [-1, 1], [1, -1]$

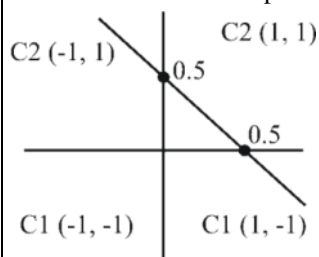
**Class C2 :**  $[1, 1]$

**The decision boundary between the two classes C1 and C2 using single perceptron is given by :**

- (a)  $x_1 - x_2 - 0.5 = 0$
- (b)  $-x_1 + x_2 - 0.5 = 0$
- (c)  $0.5(x_1 + x_2) - 1.5 = 0$
- (d)  $x_1 + x_2 - 0.5 = 0$

**Ans :** (d) Since  $x_1 + x_2 - 0.5 = 0$

Separately the points of class C2 from all the points of class C1. Hence ans is option (d)



**65. Consider a standard additive model consisting of rules of the form of**

**If  $x$  is  $A_i$  and  $y$  is  $B_i$  THEN  $z$  is  $C_i$**

**Give crisp inputs  $x = x_0$ ,  $y = y_0$  the output of the model is :**

(a)  $z = \sum_i \mu A_i(x_0) \mu B_i(y_0) \mu C_i(z)$

(b)  $z = \sum_i \mu A_i(x_0) \mu B_i(y_0)$

(c)  $z = \text{centroid} \left( \sum_i \mu A_i(x_0) \mu B_i(y_0) \mu C_i(z) \right)$

(d)  $z = \text{centroid} \left( \sum_i \mu A_i(x_0) \mu B_i(y_0) \right)$

**Ans :** (c) Standard Additive model (SAM) includes the centroid defuzzification technique. Here if  $X$  is and  $y$  is then  $z$  is given crisp inputs  $x = x_0$ ,  $y = 40$

$z = \text{centroid} (\sum_i \mu A_i(x_0) \mu B_i(y_0) \mu C_i(z))$

**66. A bell-shaped membership function is specified by three parameters (a, b, c) as follows :**

(a)  $\frac{1}{1 + \left( \frac{x-c}{a} \right)^b}$

(b)  $\frac{1}{1 + \left( \frac{x-c}{a} \right)^{2b}}$

(c)  $1 + \left( \frac{x-c}{a} \right)^b$

(d)  $1 - \left( \frac{x-c}{a} \right)^{2b}$

**Ans :** (b) A generalized bell-shaped membership function is specified by three parameters (a, b, c) as follows:

$\text{bell}(x; a, b, c) = \frac{1}{1 + \left| \frac{x-c}{a} \right|^{2b}}$

Where  $c$  and  $a$  represent the center and width, respectively of the membership function and the parameter  $b$  is usually positive.

**67. Which of the following is/are the principle components of a memory-tube display?**

- (1) Flooding gun
- (2) Collector
- (3) Phosphorus grains
- (4) Ground

**Codes**

- (a) (1) and (2)
- (b) (3) only
- (c) (4) only
- (d) All the above

**Ans :** (d) Flooding gun, collector, phosphorus grains, ground are the main components of a memory tube display. So answer is option (d)

**68. Which of the following step is/are not required for analog to digital conversion?**

- (1) Sensing
- (2) Conversion
- (3) Amplification
- (4) Conditioning
- (5) Quantization

**Codes**

- (a) (1) and (2)
- (b) (3) and (4)
- (c) (1), (2) and (5)
- (d) None of the above

**Ans :** (d) The steps which are required are  
(1) Sampling, quantization, amplification, Conversation  
every option contains at least one of these 3.  
Hence option (d) is answer.

**69. Which raster locations would be chosen by Bresenham's algorithm when scan converting a line form (1, 1) to (8, 5)?**

(a)

| x | y |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |
| 4 | 3 |
| 5 | 4 |
| 6 | 4 |
| 7 | 5 |
| 8 | 6 |

(b)

| x | y |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 2 |
| 4 | 3 |
| 5 | 4 |
| 6 | 5 |
| 7 | 6 |
| 8 | 7 |

(c)

| x | y |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 2 |
| 4 | 3 |
| 5 | 3 |
| 6 | 4 |
| 7 | 4 |
| 8 | 5 |

(d)

| x | y |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 2 |
| 4 | 3 |
| 5 | 5 |
| 6 | 4 |
| 7 | 5 |
| 8 | 5 |

**Ans :** (c) Short cut method since line has to be drawn from (1, 1) to (8, 5) so choice 1, 2 are out as they are ending up with 8, 6 & 8, 7  
Now in choice 4 after point 4, 3 it is 5, 5 i.e. increment of 2 in y which is not as per rule of line of drawing.  
So we can say answer is option (c) (because such unusual pattern is not seen is this)

70. Consider a unit square centred at origin. The coordinates of the square are translated by a factor  $\left(\frac{1}{2}, 1\right)$  and rotated by an angle of  $90^\circ$ .

What shall be the coordinates of the new square?

- (a)  $\left(\frac{-1}{2}, 0\right), \left(\frac{-1}{2}, 1\right), \left(\frac{-3}{2}, 1\right), \left(\frac{-3}{2}, 0\right)$   
 (b)  $\left(\frac{-1}{2}, 0\right), \left(\frac{1}{2}, 1\right), \left(\frac{3}{2}, 1\right), \left(\frac{3}{2}, 0\right)$   
 (c)  $\left(\frac{-1}{2}, 0\right), \left(\frac{1}{2}, 1\right), \left(\frac{-3}{2}, 1\right), \left(\frac{-3}{2}, 0\right)$   
 (d)  $\left(\frac{-1}{2}, 0\right), \left(\frac{1}{2}, 1\right), \left(\frac{-3}{2}, 1\right), \left(\frac{-3}{2}, 0\right)$

**Ans :** (a) A translation applied by a factor  $(1/2, 1)$  to all four points.

$$\Rightarrow x' = x + t_x, y' = y + t_y$$

where,  $t_x = 1/2$  and  $t_y = 1$ , therefore,  $A = (1, 3/2)$ ,  $B = (0, 3/2)$ ,  $C = (0, 1/2)$ ,  $D = (1, 1/2)$

Now, rotation by angle  $90^\circ$  according to given formula:

$$x' = x \cos 90^\circ - y \sin 90^\circ$$

$$y' = x \sin 90^\circ + y \cos 90^\circ$$

**For point A:**

$$x' = 1 \cos 90^\circ - 3/2 \sin 90^\circ = -3/2$$

$$y' = 1 \sin 90^\circ + 3/2 \cos 90^\circ = 1$$

**For point B:**

$$x' = -3/2, y' = 0$$

**For point C:**

$$x' = -1/2, y' = 0$$

**For point D:**

$$x' = -1/2, y' = 1$$

71. Which of the following is/are the components of a CRT?

- (a) Electron Gun  
 (b) Control Electrode  
 (c) Focusing Electrode  
 (d) Phosphor Coated Screen

**Codes**

- (a) (1) and (4)  
 (b) (1), (2) and (4)  
 (c) (1), (2), (3) and (4)  
 (d) (1), (3) and (4)

**Ans :** (c) A CRT is an electronic tube designed to display electrical data.

The basic CRT consist of four major components.

- (1) Electron gun  
 (2) Focusing & Accelerating A nodes  
 (3) Horizontal & vertical deflection plates  
 (4) Evacuated glass enclose

72. Which one of the following statements is incorrect?

- (a) Pareto analysis is a statistical method used for analyzing causes, and is one of the primary tools for quality management
- (b) Reliability of a software specifies the probability of failure-free operation of that software for a given time duration
- (c) The reliability of a system can also be specified as the mean time to failure (MTTF)
- (d) In white-box testing, the test cases are decided from the specifications or the requirements

**Ans :** (d) White box testing are connected with codes in block box testing the test case are decided from the specification of the requirement.

So answer is option (d)

**73. Consider a language A defined over the alphabet  $\Sigma = \{0,1\}$  as  $A = \{0^{[n/2]}1^n : n \geq 0\}$ .**

**The expression  $[n/2]$  means the floor of  $n/2$ , or what you get by rounding  $n/2$  down to the nearest integer.**

**Which of the following is not an example of a string in A?**

- (a) 011
- (b) 0111
- (c) 0011
- (d) 001111

**Ans :** (c) If  $n = 2$

$[n/2] = [1] = 1$  i.e. language is : 011

If  $n = 3$

$[n/2] = [1.5] = 1$  i.e. language is : 0111

If  $n = 4$

$[n/2] = [2] = 2$  i.e. language is : 00111

Hence option (a) is possible, (b) is possible, (d) is possible but (c) is not possible

**74. Which one of the following statements, related to the requirements phase in software Engineering, is incorrect?**

- (a) "Requirement validation" is one of the activities in the requirements phase
- (b) "Prototyping" is one of the methods for requirement analysis
- (c) "Modeling-oriented approach" is one of the methods for specifying the functional specifications
- (d) "Function points" is one of the most commonly used size metric for requirements

**Ans :** (c)

**75. .... tag is an extension to HTML that can enclose any number of Javascript statements.**

- (a) <SCRIPT>
- (b) <BODY>
- (c) <HEAD>
- (d) <TITLE>

**Ans :** (a) One approach for specifying functional requirement is through such methods as a system content diagram or a use-case model.